

Operators

A Few Thoughts

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Current Topics of Interest

- Mixture Of Experts (GroupedMatMul)
- KV Cache Quantization (& Quantized Attention)
 - FlexAttention
- Paged Attention
- Multihead Latent Attention (MLA)
- Packed Attention
- Speculative Decoding
 - Only minor impact on ops per se (need for non-causal mask)
- RotaryEmbedding 2D (for vision models)

Need For Fused Ops

- Primitive ops in ONNX suffice to *express* most new ideas
- Fused ops are critical for *performance* (an implementation concern)
- Can backends (compilers) handle the optimization? What's needed from the onnx models to help?
 - Input from compiler-writers would help

Challenges, Tradeoffs, and Options

- Runtime code-generation forbidden/hard (iOS, Android)
- Compiler can increase binary size (MLIR/LLVM)
- Compilation time can be high
 - AOT Compilation and Caching
- Custom compilation (of a specific code-fragment) can exploit more optimizations than generic compilation
 - “Custom compilation” == authoring a kernel for a specific (fused) op
 - Can be done by human developer ... or an ai agent
 - Fused ops will continue to be important
 - Identifies a common pattern of computation worth optimizing heavily AOT, one that can be reused across different models

FlexAttention

- FlexAttention is a higher-order op with subgraph-attributes
 - An op template
- Efficient implementation requires compilation support: an interpreted implementation doesn't serve the purpose of a fused op
- It helps avoid the need for graph-pattern-matching to identify subgraphs to be fused
- Will compiled version be as efficient as specialized kernels for common cases (instances)?
- Implementations matching subgraphs against patterns?
- Appropriate for research/experimentation; placed in experimental domain

Scan Op Extensions

- Scan is an operator that represents an iteration over a collection of (same-shape) tensors represented as a single tensor (along a dimension)
- Proposed extension to represent iteration over a collection of (variable length) tensors represented as a single tensor (by packing)
 - Example: Packed Attention
- Works well without fusion in some settings (eg: CPU).
 - May need compilation support in other settings (eg: GPU)
- May be useful to extend to represent Parallel Scan